

DATABASE SYSTEMS

CHAP. 6: Query processing and Optimization Techniques

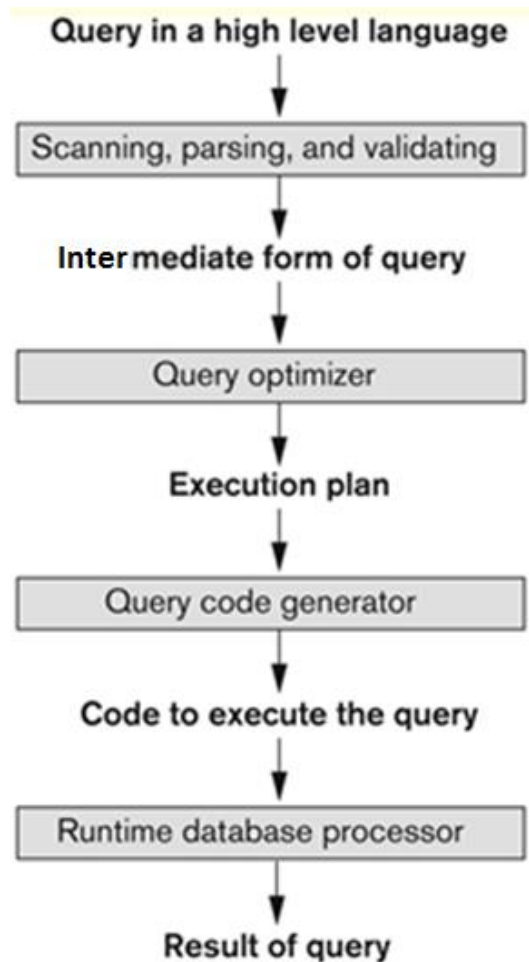
Processing a Query

- Tasks in processing a high-level query
 - Scanner scans the query and identifies the language tokens
 - Parser checks the syntax of the query
 - The query is validated by checking that all attribute names and relation names are valid

Processing a Query(Cont.)

- An intermediate internal representation for the query is created (query tree or query graph)
 - Query execution strategy is developed
 - Query optimizer produces an execution plan
 - Code generator generates the object code
 - Runtime database processor executes the code
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- Query processing and query optimization

Typical steps when processing a high-level query



Translating SQL queries into RA

- SQL query is translated into an equivalent ***extended*** relational algebra expression --- represented as a query tree
- In order to transform a given query into a query tree, the query is decomposed into ***query blocks***
 - **Query block:**
 - The basic unit that can be translated into the algebraic operators and optimized.
 - A query block contains a single SELECT-FROM-WHERE expression, as well as GROUP BY and HAVING clause if these are part of the block.
- The query optimizer chooses an execution plan for each block

Translating SQL queries into RA (Cont.)

- For example, the following compound query:

```
SELECT      Lname, Fname
FROM EMPLOYEE
WHERE Salary > ( SELECT MAX(Salary)
                  FROM EMPLOYEE
                  WHERE Dno = 5      )
```

- Can be decomposed into two blocks:

```
SELECT FNAME, LNAME
FROM EMPLOYEE
WHERE SALARY > c
```

- And

```
SELECT MAX(SALARY)
FROM EMPLOYEE
WHERE DNO=5
```

- Where 'c' is the result returned from the inner query block.

Translating SQL queries into RA (Cont.)

- The inner query block (which need to be calculated first) could be translated into the expression

$\delta \langle \text{MAX SALARY} \rangle (\sigma \langle \text{DNO}=5 \rangle (\text{EMPLOYEE}))$

- And the outer block into the expression

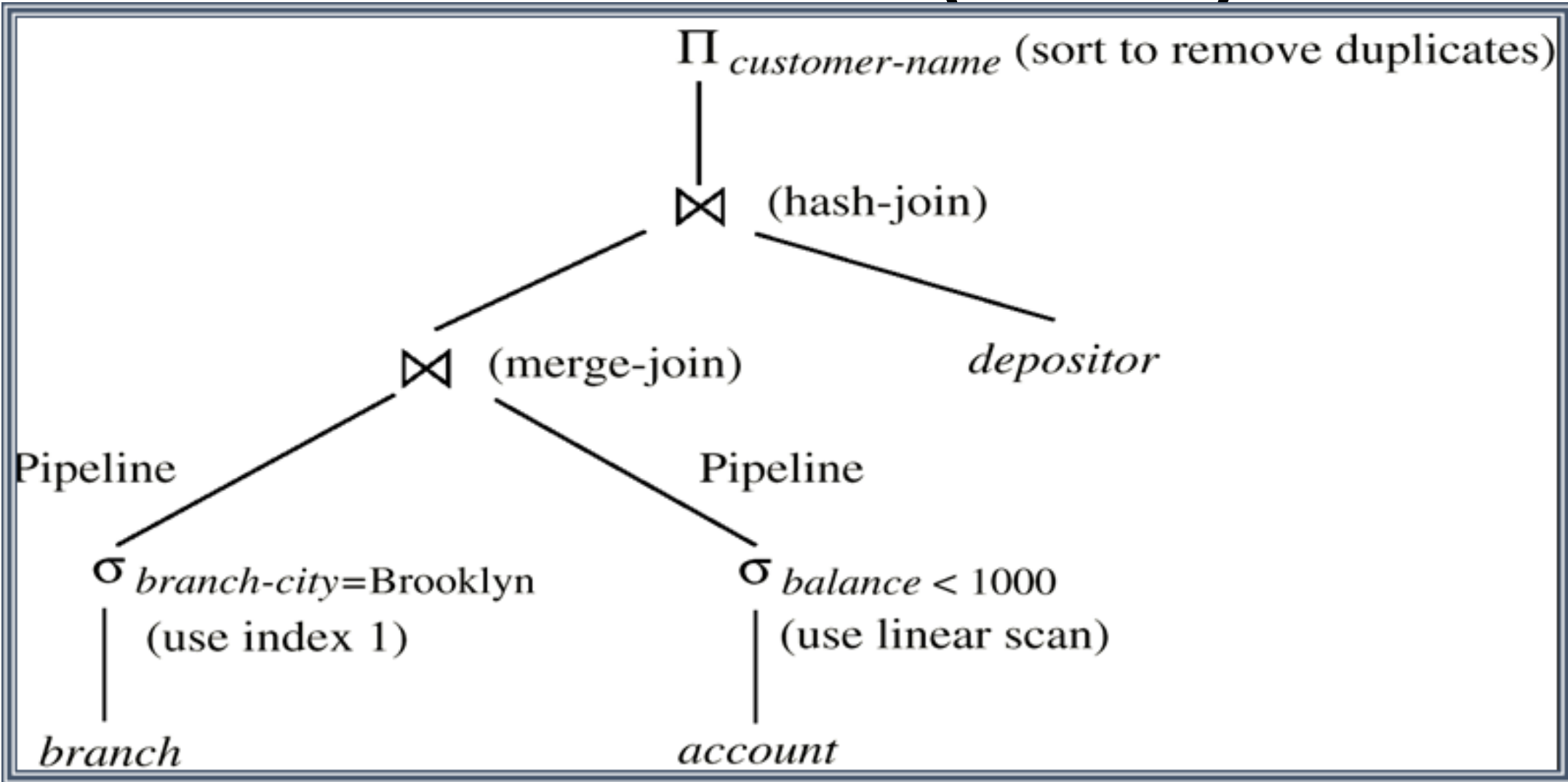
$\pi \langle \text{FNAME, LNAME} \rangle (\sigma \langle \text{SALARY} > c (\text{EMPLOYEE}))$

- The query optimizer would then chooses an execution plan for each block.

Evaluation Plan

- An evaluation plan defines exactly what algorithm is used for each operation, and how the execution of the operations is coordinated.

Evaluation Plan(Cont.)



Initial canonical query tree

- Given the following tables:

Book (access#, title)

Member (ticket#, name)

Loan(loanedbook,loanedto)

- And the SQL query:

Select member.name

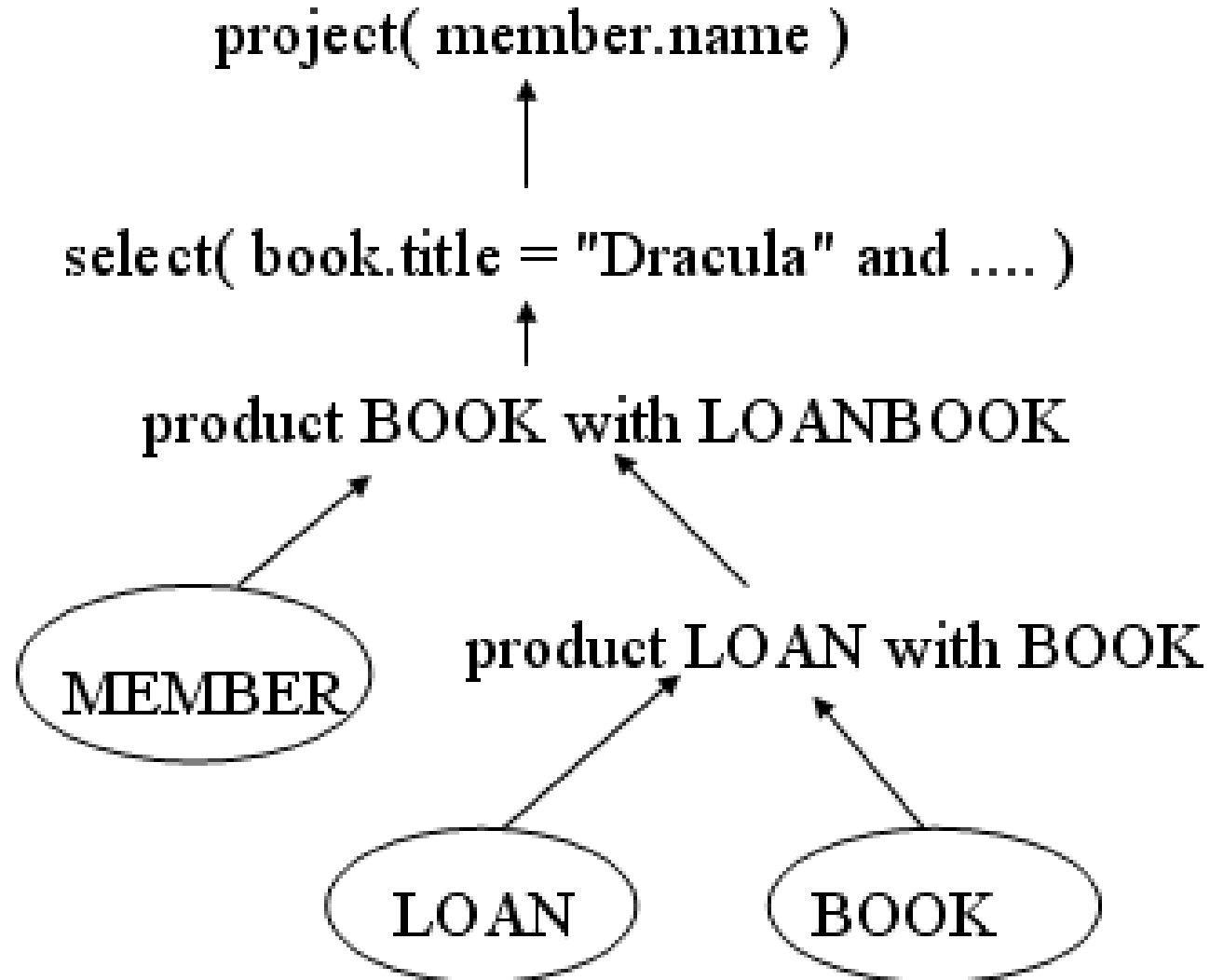
From book, loan, member

where book.title = "dracula"

and member.ticket# = loan.loanedto

and loan.loanedbook = book.access#

Initial canonical query tree (Cont.)



Query Execution

- An execution plan for a relational algebra query consists of a combination of the relational algebra query tree and information about the access methods to be used for each relation as well as the methods to be used in computing the relational operators stored in the tree.

Query optimization

- Amongst all equivalent evaluation plans choose the one with lowest cost
- The process of choosing a suitable execution strategy for processing a query.

Cost estimations

- ***Cost components for query execution***
 - Access to secondary storage (searching, reading, writing blocks)
 - Storage cost (intermediate files)
 - Computational cost (searching, sorting, merging in main memory)
 - Memory usage cost (memory buffers)
 - Communication cost (distributed dbs)